

SWASTIK RAJ

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SKILLS

Programming Languages: C++, Python, Java, JavaScript, HTML, CSS

Software Frameworks and Tools: Node.js, Express.js, FastAPI, Rest APIs, Docker, Git, GitHub

Database: SQL, MongoDB, PostgreSQL

Computer Science: Data Structures and Algorithms, OOPs

EXPERIENCE

UNIFIED MENTOR

JUNE 2025 - JULY 2025

WEB DEVELOPER INTERN (REMOTE)

Mumbai, INDIA

- Developed and deployed responsive web applications using **HTML, CSS, and JavaScript**, optimizing UI performance and enhancing user experience.
- Engineered backend services and **RESTful APIs** using **Node.js** and **Express.js**, enabling **client-server** communication.
- Designed and managed **PostgreSQL** relational databases with optimized schema structures and query handling.

PROJECTS

Emotion Voice Assistant Website

Tech Stack: HTML, CSS, JavaScript, Node.js, Express.js, Gemini API

- Built full-stack emotion-aware voice assistant using **Node.js, Python FastAPI, PyTorch** achieving **<500ms API latency** with **7-class emotion detection** from real-time webcam feeds.
- Integrated **Google Gemini API** for intelligent conversational responses with **real-time speech-to-text (Web Speech API)** and **text-to-speech synthesis** having emotion context awareness with multi-language microservices architecture combining **PyTorch CNN model** with **<100ms** frame processing.

Secure Webhook Ingestion API

Tech Stack: Python, FastAPI, SQLite, Docker, Docker Compose, Uvicorn, Webhook

- Built **RESTful API** backend using FastAPI and Python with **HMAC-SHA256 webhook** authentication, storing and validating **100+ messages** daily with **idempotent processing**.
- Containerized microservice deployment with **Docker and Uvicorn**, implementing **4 core endpoints** with automated startup initialization and health monitoring.
- Engineered SQLite database schema with message deduplication using primary keys, implemented **Pydantic validation** ensuring **100 percent** data integrity across **API payload handling**.

Library Management System

Tech Stack: React, Tailwind, Node.js, Express.js, MongoDB Atlas, bcrypt.js, Axios, Groq API

- Full-stack **MERN** application – **React.js** frontend with **Node.js/Express** backend and **MongoDB**, featuring **JWT-based authentication** and **error handling middleware**.
- Book Management Borrowing System – Developed complete book catalog with **real-time inventory tracking, automated 7-day lending workflow, and availability management**.
- AI-Powered Chatbot – Integrated **Groq AI API** for intelligent library assistant providing book recommendations and user support.

EDUCATION

Amity University Mumbai – B.Tech in Computer Science and Engineering (GPA: 8.65) [Aug 2023 - Jul 2027]

Eklavya Educational Complex – Class 12th CBSE [2022]

ACHIEVEMENT

- **LeetCode:** Solved 100+ Algorithmic and Data Structure Problems